Integrating Models and Agent-based systems

Dynamic Modelling for Human-centered Systems

Chapter 13

Course outline

- Week 1: what is dynamic modelling?
- Week 2: modelling social processes
- Week 3: modelling behaviour
- Week 4: domain models as basis for intelligent applications
- **Week 5: agent-based modelling**
- Week 6: analysis and support models
- Week 7: wrap-up

domain models

Overview of today's lecture

1. **Brief recap of models as basis for intelligent applications**
2. Agent-based modelling
3. Ontologies for agent-based models

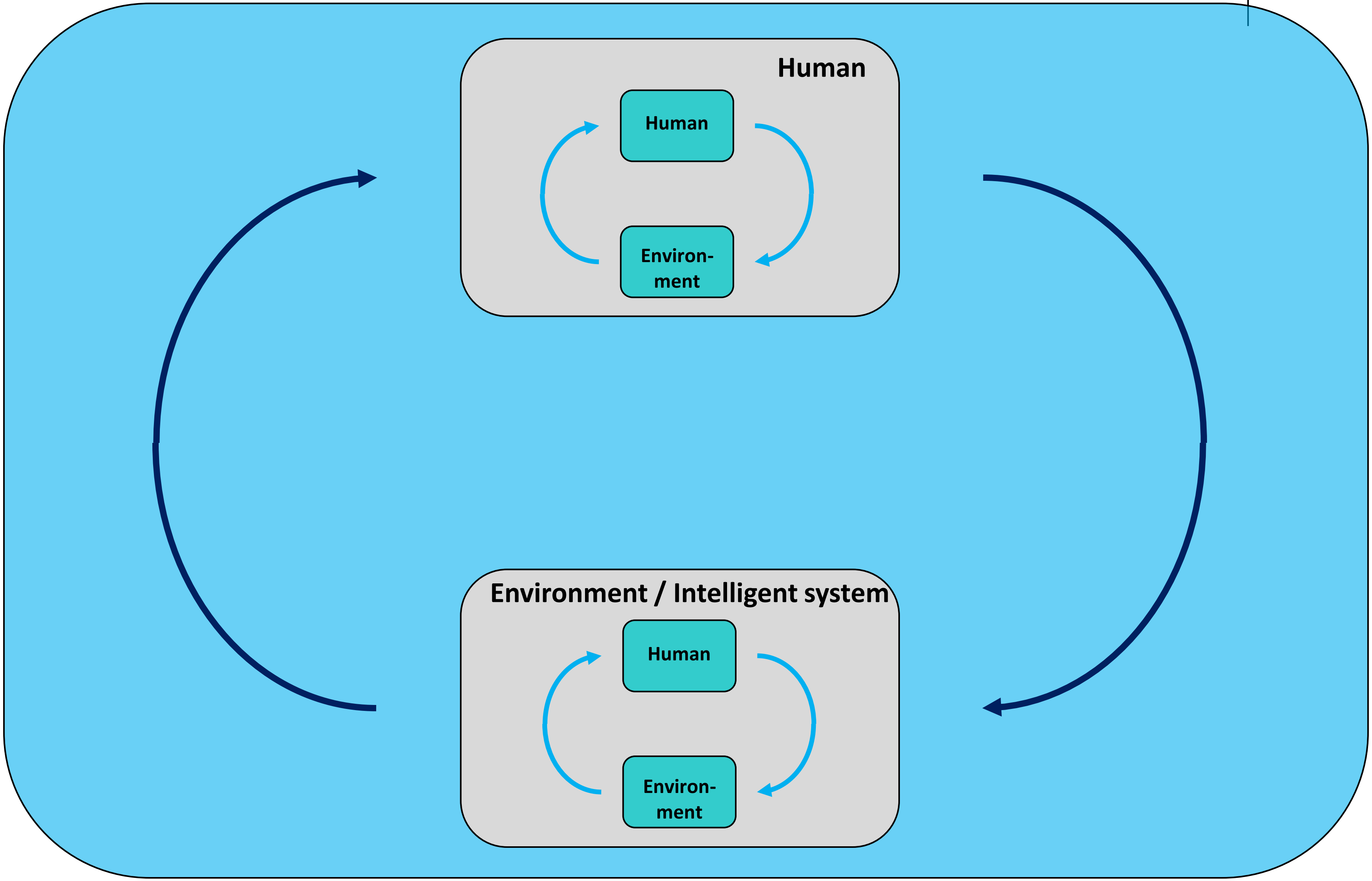
Building intelligent systems human-centered systems

- Intelligent human-centered systems:
  - interact and cooperate with users
  - adapt to needs and preferences of users
- What is needed?

→ **knowledge** about humans and how they function

- In this course:
  - incorporate **dynamic models** about human functioning

Ambient Intelligence: Global Idea



Two types of reasoning

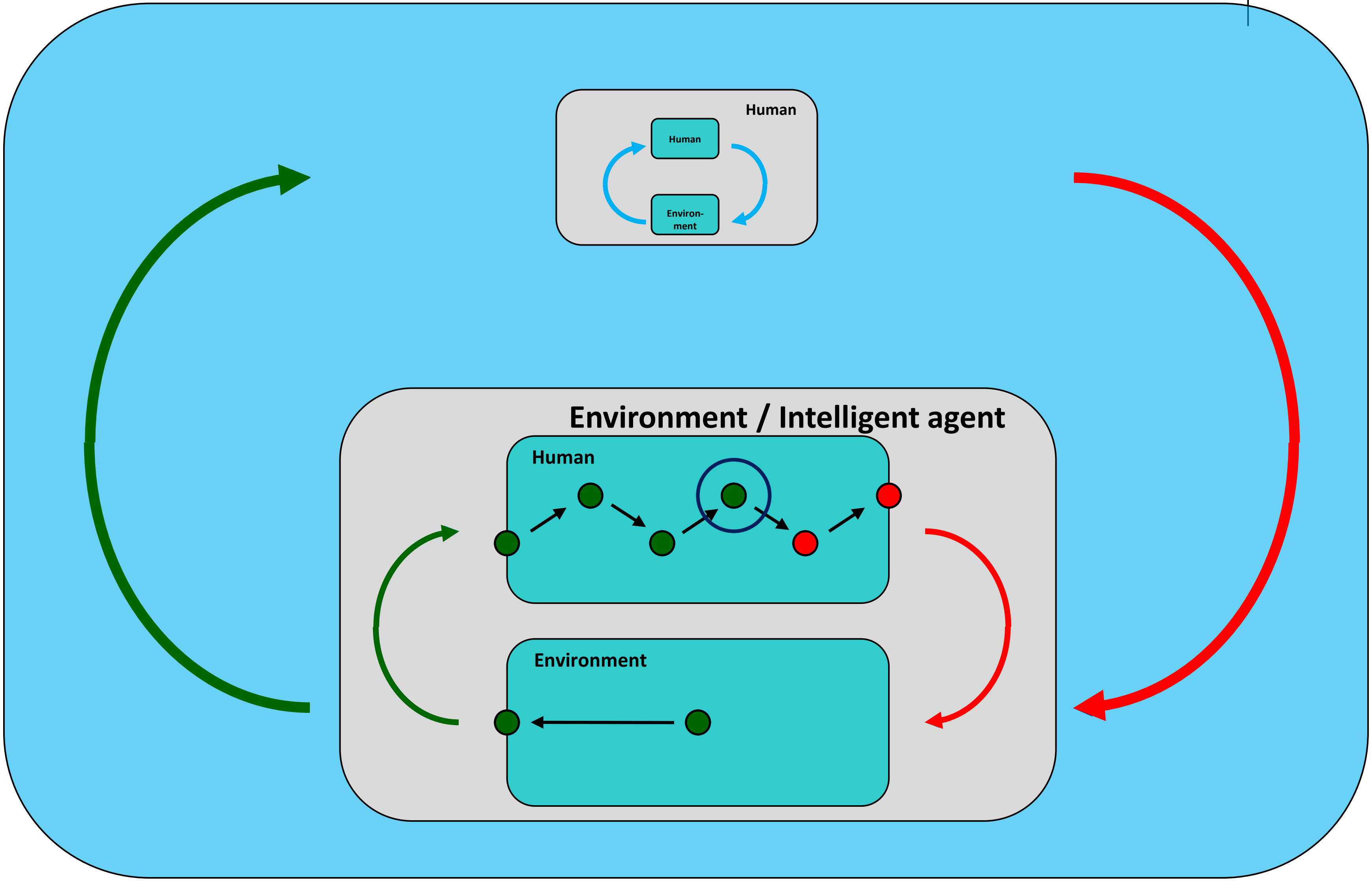
- What is going on?
  - making **assessments** of the current situation

- What will happen if?
  - **predicting** the effect of possible interventions

Two types of reasoning

- Making **assessments** of the current situation
  - ‘This user has a sad face and has used negative words over the last 2 hours, he is probably depressed’
  - ‘This driver is blinking with her eyes all the time, she is probably getting tired’
- **Predicting** the effect of possible interventions
  - ‘If the user goes for a walk in the park, he might feel better’
  - ‘If the driver takes a break of 30 minutes, she will recover’

Using Models within intelligent systems



Position within Artificial Intelligence

- Two main perspectives in AI:
  - **Top-down** (knowledge driven) AI
  - **Bottom-up** (data driven) AI: machine learning

- This course focuses on the **first** perspective
  - per definition ‘Explainable AI’

Overview of today's lecture

1. Brief recap of models as basis for intelligent applications
2. **Agent-based modelling**
3. Ontologies for agent-based models

What is an Agent?

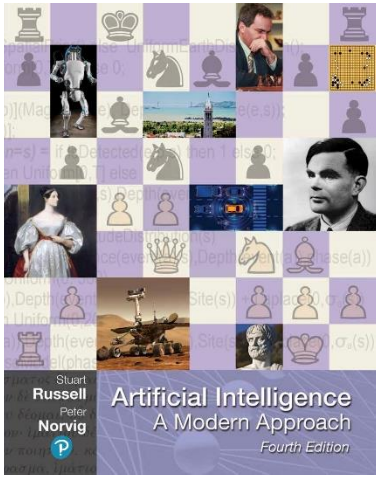
- A term used in various disciplines
  - philosophy
  - social sciences
- Latin: agere = to do
  - an agent is an entity that does something
  - being an agent = agency
- Agents in AI:
  - not only humans, but also computer (programs)

Agent-based modelling

- Agent-oriented development is **paradigm** of looking at software development
  - not one large program
  - small pieces of software that have their own goals and communicate with others
- Useful paradigm for building intelligent systems
  - both **humans** and **computer systems** can be considered as interacting agents
  - computer systems can use simulations of human behaviour / thinking for own actions

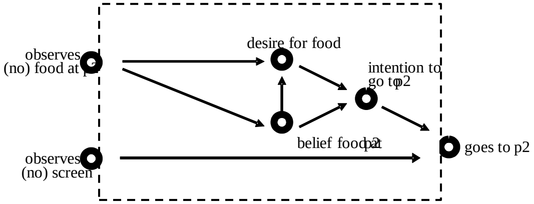
Agents in AI

- "Standard model" in AI (Russell and Norvig, 2022):
  - rational agents
- Objective of AI programs is to act in the right way
  - assumes that explicit task is given



Agents and models

- Up to now:
  - models of animals / humans
  - the entity that we modelled can be seen as an **agent**
  - mostly implicit



- From now on:
  - make the agent explicit
    - observed(**mouse1**, food_at(p2))
  - also systems can be agents
    - observed(**smartwatch1**, hearthrate(**person1**, 83))

Integrating Models within Intelligent Systems

- Examples of **intelligent systems**:
  - intelligent cars
  - smart homes
  - search bots / recommender systems on Internet
  - robots

  → called: **agents**
- **Actors** in our models
  - mouse
  - Tom Thumb
  - humans

  → called: **agents**

Why using agent paradigm?

- Five ongoing trends within computing:
  - *ubiquity*
    (processing power is everywhere)
  - *interconnection*
    (computer systems are networked and distributed)
  - *intelligence*
    (computers solve increasingly complex tasks)
  - *delegation*
    (we provide more and more control to computers)
  - *human-orientation*
    (machines are increasingly being treated as humans)

Ubiquity

- Computing capability is becoming *cheaper*
- Processing power is introduced *everywhere*
- ‘*Intelligence*’ becomes ubiquitous



Interconnection

- Computer systems are *networked*
- Most obvious example: the *Internet*
- Computing = primarily a process of *interaction*?

Intelligence

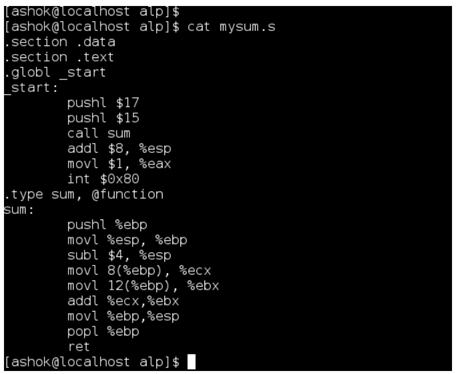
- The *complexity* of tasks that we are capable of automating and delegating to computers has grown steadily

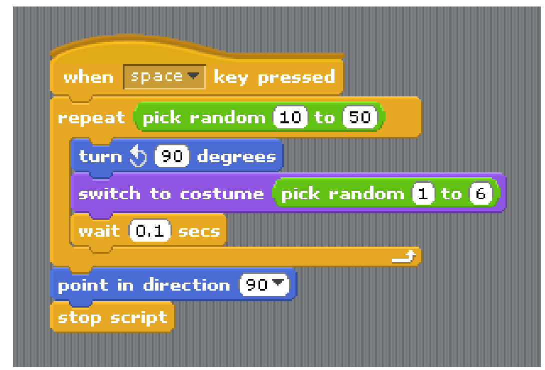
Delegation

- Computers are doing more for us (without our intervention)
- We are *giving control* to computers, even in safety critical tasks

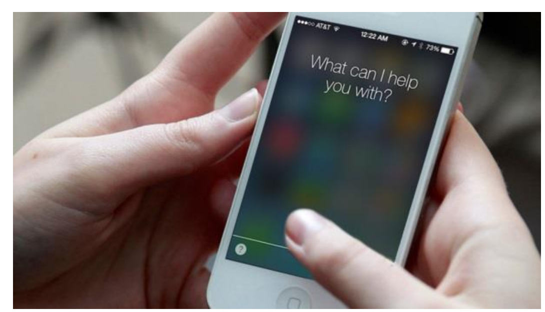
Human Orientation

- Movement from machine-oriented toward more user-friendly programming





- Programmers (and users!) relate to the machine differently



What does this mean for computer systems?

- *Delegation* and *intelligence* imply the need to build computer systems that can act effectively **on our behalf**
- This implies:
  - the ability of computer systems to act **independently**
  - the ability of computer systems to act in a way that **represents our best interests** while interacting with other humans or systems

→ agents

Definition of Agency

"An agent is a computer system that is situated in some environment, and that is capable of autonomous action in this environment in order to achieve its delegated goals"

- Behaviour: sense - decide - act - …
- However: applies also to very 'dumb' agents

Characteristics of Intelligent Agents

- Autonomous
  - Able to control its own processes independently
- Reactive
  - Able to receive information from environment and respond
- Pro-active
  - Able to take initiative to work systematically towards your goals
- Social
  - Able to communicate, coordinate, negotiate and cooperate

Question

- Can you think of some examples of natural and artificial intelligent agents?
- Why do these satisfy the definition?

Examples of Natural Agents

- Humans
  - Travel agent, traffic agent, insurance agent, secret service agent
- Primates
- Cats
- Bacteria
- Plants
- …

Examples of Artificial Agents

- Thermostat
- Intelligent car
- Smart home
- Robot
- Game character
- Search engine
- …

Summary

- Agent paradigm:
  - small pieces of software that have their own goals and communicate with others
- Useful when describing interaction between intelligent systems and humans

- Has standard vocabulary
  - we have already seen a bit of this

    observed(agent, food_at_position(p2))

Overview of today's lecture

1. Brief recap of models as basis for intelligent applications

2. Agent-based modelling

3. **Ontologies for agent-based models**

What is an ontology?

- Currently: standard vocabulary of terms to be used in models
  - our “dictionary” to use
- Later: formal categorisation of terms, their relation and meaning
  - logical structure

Modelling Intelligent Agents

We introduce some primitives:

- External Agent Concepts
- Interface Agent Concepts
  - Input Concepts
  - Output Concepts
- Internal Agent Concepts

External Agent Concepts

- Processes and States in the External World
  - The sun is shining
  - It is raining
  - Agent A is at location X
  - …

Interface Agent Concepts

- Interaction with the External World
  - Observation
    - Passive (i.e. without taking initiative)
    - Active (i.e. actively initiated and controlled)
  - Execution of Actions

    (to change the state of the environment)

- Communication with Other Agents
  - Outgoing (i.e. sending)
  - Incoming (i.e. receiving)

Example 1a

- For a Thermostat, can you think of any:
  - Observations?
    - Passive?
    - Active?
  - Actions?
  - Communications?
    - Outgoing?
    - Incoming?

Internal Agent Concepts

- Which concepts did we use for describing pro-active behavior?

- Beliefs
  - about the world
- Desires
  - goals the agent wants to achieve
- Intentions
  - plans of action to achieve its goals

Internal Agent Concepts

- World Model
  - about the state of the external world
- Agent Model
  - about the state of other agents around
- Self Model
  - about its own characteristics, state, and behaviour
- History
  - of world, agents, and self
- Desires
  - goals the agent wants to achieve
- Intentions
  - plans of action to achieve its goals

Beliefs

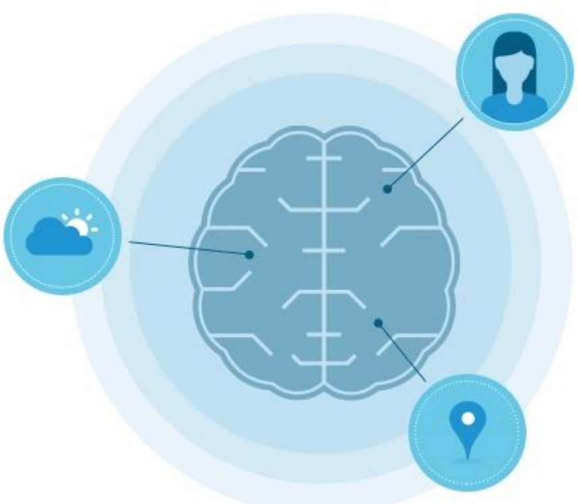
Example 1b

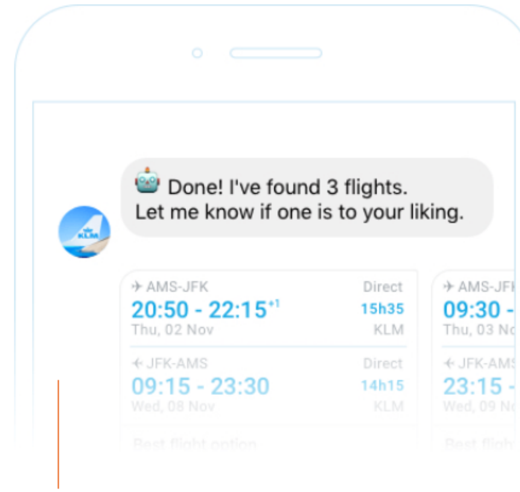
- For a Thermostat, can you think of any:
  - World Model?
  - Agent Model?
  - Self Model?
  - History?
  - Desires?
  - Intentions?

Example 2 – Blue Bot KLM's smart assistant

Welcome to the family

Meet BB (that's short for Blue Bot), a member of our KLM service family. Just like her 300 human colleagues, she's dedicated to provide you with the best possible service. You can meet BB on Messenger for booking a flight and wherever the Google Assistant is, for instance on Google Home or your smartphone, for finding a destination, booking a flight and packing your bag.





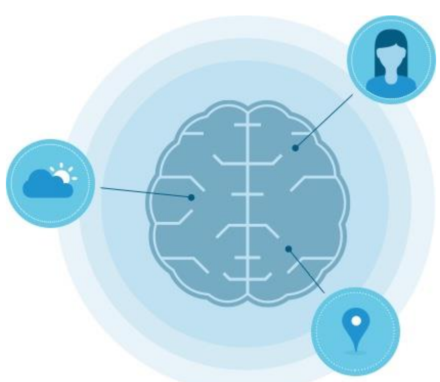
Book a ticket

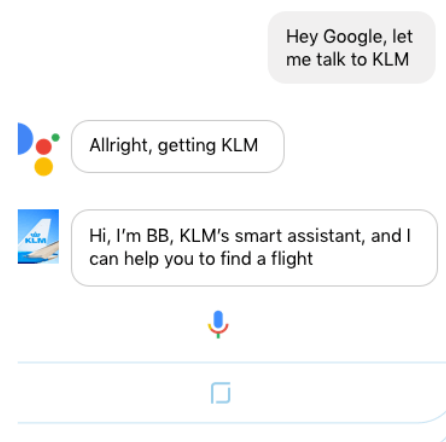
If you'd like to book a ticket, there's another easy way to do so. On Messenger, BB can help you with booking your ticket. It's smart and easy!

Find BB on Messenger

Example 2

- For an **internet agent**, can you think of any:
  - Observations?
    - Passive?
    - Active?
  - Actions?
  - Communications?
    - Outgoing?
    - Incoming?
  - World Model?
  - Agent Model?
  - Self Model?
  - History?
  - Desires?
  - Intentions?





Primitive Agent Concepts

- (External Agent Concepts)
- Interface Agent Concepts
  - Observations
  - Actions
  - Communications
- Internal Agent Concepts
  - World Model, Agent Model, Self Model
  - History
  - Desires, Intentions

- Next step: formalisation of these concepts

Interface Agent Concepts - Formal Ontology

concept	*predicate*
passive observation	**observed : AGENT x INFO_ELEMENT**
active observation	**to_be_observed : AGENT x INFO_ELEMENT**
action	**performed : AGENT x ACTION**
incoming communication	**communicated_from_to :** **INFO_ELEMENT x AGENT x AGENT**
outgoing communication	**to_be_communicated_from_to :** **INFO_ELEMENT x AGENT x AGENT**

Internal Agent Concepts - Formal Ontology

concept	***predicate***
belief	**belief : AGENT x INFO_ELEMENT**
desire	**desire : AGENT x INFO_ELEMENT**
intention	**proposed : AGENT x ACTION**

Can be used for World Model, Agent Model, Self Model, and History!

Case Study - Depression Support

In modern society, depression is a mental illness with a significant impact on the patient suffering from a depression, the direct environment of the patient, but also the society in general. A problem with treating people suffering from a depression is that it is quite a step for a person suffering from a depression to go to a general practitioner, or a psychologist, explain his/her problems, and receive appropriate treatment. As a consequence, too few people receive treatment, and remain in their depression for an unnecessary amount of time. In order to improve these matters, it is envisioned to develop an intelligent support system for people suffering from a depression. This system is briefly described below.

PASSIVE OBSERVATIONS?

Case Study - Depres

The intelligent support system is meant to monitor the patient, provide therapy, and give dedicated feedback to the patient. In order to do so, the system continuously monitors the behavior of the patient. It hereby monitors the activity of the patients via sensors (e.g. motion sensors, heart rate sensors, but also by looking at GPS locations), but also actively asks the patient questions what the patient is currently doing. Next to the activities, the system also monitors the mood of the patient, this is done by asking the patient how he is currently feeling. The providing of therapy consists of giving homework assignments to the patient, and showing relevant information to the patient. All of this is done in a joint effort between patient and system to come to a rapid recovery. The final element is to provide feedback to the patient, which is simply done by sending text messages, or displaying some graphical information (e.g. the mood over the past week).

ACTIVE OBSERVATIONS?

Case Study - Depres

The intelligent support system is meant to monitor the patient, provide therapy, and give dedicated feedback to the patient. In order to do so, the system continuously monitors the behavior of the patient. It hereby monitors the activity of the patients via sensors (e.g. motion sensors, heart rate sensors, but also by looking at GPS locations), but also actively asks the patient questions what the patient is currently doing. Next to the activities, the system also monitors the mood of the patient, this is done by asking the patient how he is currently feeling. The providing of therapy consists of giving homework assignments to the patient, and showing relevant information to the patient. All of this is done in a joint effort between patient and system to come to a rapid recovery. The final element is to provide feedback to the patient, which is simply done by sending text messages, or displaying some graphical information (e.g. the mood over the past week).

Case Study - Depres

OUTGOING COMMUNICATIONS?

The intelligent support system is meant to monitor the patient, provide therapy, and give dedicated feedback to the patient. In order to do so, the system continuously monitors the behavior of the patient. It hereby monitors the activity of the patients via sensors (e.g. motion sensors, heart rate sensors, but also by looking at GPS locations), but also actively asks the patient questions what the patient is currently doing. Next to the activities, the system also monitors the mood of the patient, this is done by asking the patient how he is currently feeling. The providing of therapy consists of giving homework assignments to the patient, and showing relevant information to the patient. All of this is done in a joint effort between patient and system to come to a rapid recovery. The final element is to provide feedback to the patient, which is simply done by sending text messages, or displaying some graphical information (e.g. the mood over the past week).

Case Study - Depres

INTERNAL STATES?

The intelligent support system is meant to monitor the patient, provide therapy, and give dedicated feedback to the patient. In order to do so, the system continuously monitors the behavior of the patient. It hereby monitors the activity of the patients via sensors (e.g. motion sensors, heart rate sensors, but also by looking at GPS locations), but also actively asks the patient questions what the patient is currently doing. Next to the activities, the system also monitors the mood of the patient, this is done by asking the patient how he is currently feeling. The providing of therapy consists of giving homework assignments to the patient, and showing relevant information to the patient. All of this is done in a joint effort between patient and system to come to a rapid recovery. The final element is to provide feedback to the patient, which is simply done by sending text messages, or displaying some graphical information (e.g. the mood over the past week).

desire(agent, patient_mood(high))

Case Study - Depression Support

The intelligent support system is meant to monitor the patient, provide therapy, and give dedicated feedback to the patient. In order to do so, the system continuously monitors the behavior

observed(agent, heart_rate(80))

of the patient: it thereby monitors the activity of the patients via sensors (e.g. motion sensors, heart rate sensors, but also by looking at GPS locations), but also actively asks the patient questions what the patient is currently

belief(agent, patient_mood(low))

to_be_observed(agent, patient_activity)

also monitors the mood of the patient, this is done by asking the patient how he is currently feeling. The providing of therapy consists of giving homework assignments to the patient, and showing relevant information to the patient. All of this is done in a joint effort between patient and system to come to a rapid re

to_be_communicated_from_to(agent, patient, mood_level(X))

ment is to provide feedback to the patient, done by sending text messages, or displaying some graphical information (e.g. the mood over the past week).

Overview of Modelling Concepts

- (External Agent Concepts)
- Interface Agent Concepts
  - Observations, Actions, Communication, …
- Internal Agent Concepts
  - Beliefs, Desires, Intentions, …
- Domain-Specific Concepts
  - Design your own...

Conclusion - 1

- Smart applications can be build by:
  - integrating models of human functioning within applications
  - goal: application that can reason about humans

    → **assessment** of the situation

    → decision on **support**

- Agent:
  - capable of autonomous action in environment to achieve goals

- Intelligent Agent:
  - reactive, pro-active, social
  - also intentional properties

Conclusion - 2

- Primitive Agent Concepts:
  - external
  - interface
  - internal
  - specific concepts for Integrative Modelling

- Formal Ontology:
  - standard predicates and sorts for agent concepts